Chapter 11

Field Effect Transistors (JFET)

11.1 Introduction

While the Bipolar Junction Transistor (BJT) is a powerful device, it has a significant drawback: it is **Current Controlled**. The base current (I_B) controls the output current (I_C). This means a BJT always draws some current from the signal source, resulting in a relatively low input impedance (Z_{in}).

The **Field Effect Transistor (FET)** solves this. It is a **Voltage Controlled** device. The voltage at the Gate terminal controls the current flowing from Source to Drain using an electric field, with virtually **zero input current**.

BJT vs. FET

- **BJT:** Bipolar (Electrons & Holes), Current Controlled ($I_C = \beta I_B$), Low Input Impedance.
- **FET:** Unipolar (Electrons OR Holes), Voltage Controlled (I_D func of V_{GS}), High Input Impedance ($M\Omega$).

Learning Objectives

By the end of this chapter, you should be able to:

- Describe the construction of N-Channel and P-Channel JFETs.
- Explain the ”Pinch-Off” mechanism using depletion regions.
- Interpret the Drain and Transfer characteristic curves.
- Apply Shockley’s Equation to calculate Drain Current.
- Compare JFETs with BJTs in terms of impedance and gain.

11.2 JFET Construction

The **Junction Field Effect Transistor (JFET)** is the simplest type of FET. It essentially consists of a conductive bar (the Channel) with a control terminal (the Gate).

11.2.1 N-Channel JFET

- **Channel:** A bar of N-type semiconductor material.
- **Terminals:**
  - **Source (S):** Where electrons enter the channel.
  - **Drain (D):** Where electrons leave the channel.
  - **Gate (G):** Two regions of P-type material diffused into the middle of the N-channel.

The Gate-Channel junction creates a **PN Junction**. Under normal operation, this junction is always **Reverse Biased**, creating a depletion region that encroaches into the channel.

11.3 Principle of Operation

The JFET operates like a water tap.

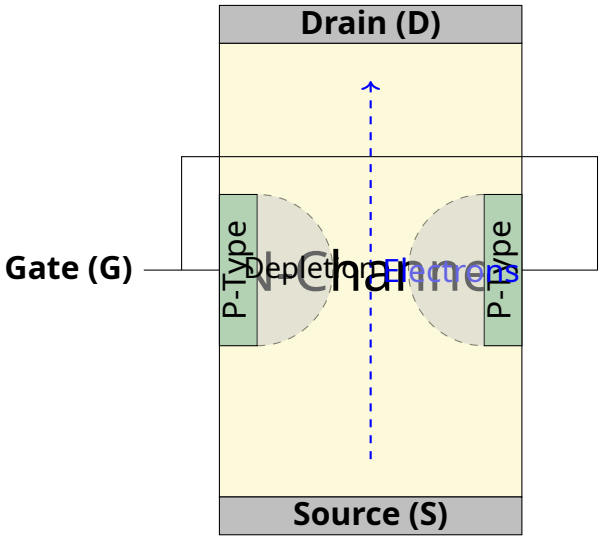


Figure 11.1: Physical Structure of an N-Channel JFET.

- **Source/Drain Voltage (V_{DS}):** The water pressure (pump).
- **Gate Voltage (V_{GS}):** The tap handle (valve).
- **Drain Current (I_D):** The water flow.

11.3.1 Case 1: $V_{GS} = 0$V (Gate Shorted to Source)

Suppose $V_{GS} = 0$V and we apply a positive voltage V_{DS} at the Drain.

- Electrons flow from Source to Drain.
- The channel is wide open.
- Maximum current flows. This maximum saturation current is denoted as $\mathbf{I_{DSS}}$ (Drain-Source Saturation Current).

11.3.2 Case 2: V_{GS} is Negative (Reverse Bias)

Now, we apply a negative voltage to the Gate (e.g., $V_{GS} = -2$V).

1. The Gate-Channel PN junction becomes reverse biased.
2. The depletion region widens.
3. Since the depletion region is devoid of carriers, it effectively "squeezes" the N-channel, making it narrower.

4. Resistance increases $\rightarrow$ Drain Current (I_D) decreases.

11.3.3 Pinch-Off Voltage (V_P)

As we make the Gate more negative, a point is reached where the two depletion regions meet (touch) in the center of the channel.

- This completely blocks the path for electrons.
- The current I_D drops to zero (or effectively zero leakage).
- The negative voltage at which the current drops to zero is called the **Pinch-Off Voltage (V_P)**.
- For N-channel, V_P is a negative number (e.g., -4V).

11.4 JFET Characteristics

11.4.1 Transfer Characteristics

Unlike the BJT, where output current is linear with input current, the JFET output current is related to the input voltage by the **Shockley Equation**:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 \tag{11.1}$$

Where:

- I_D: Drain Current.
- I_{DSS}: Saturation Current (Max current when $V_{GS} = 0$).
- V_{GS}: Gate-to-Source Voltage (Control variable).
- V_P: Pinch-off Voltage (Device constant).

This relationship creates a **Parabolic Curve**.

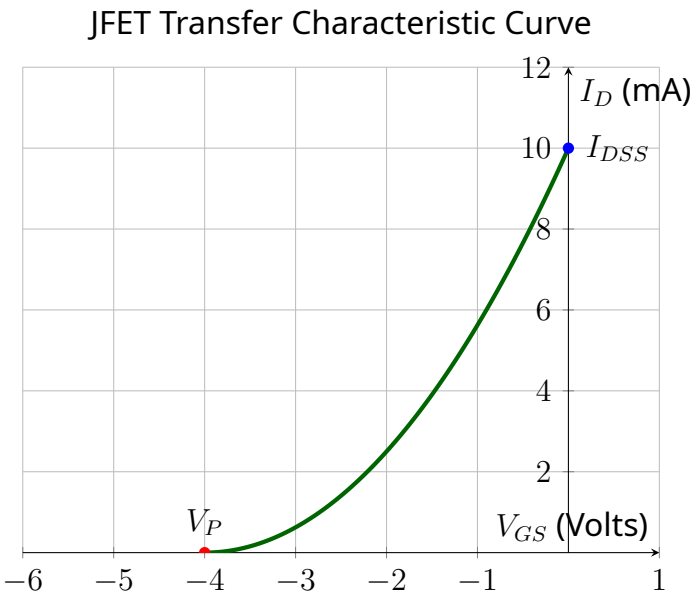


Figure 11.2: Transfer Curve for an N-Channel JFET ($I_{DSS} = 10$mA, $V_P = -4$V).

11.5 JFET Symbols

The schematic symbols distinguish between N-Channel and P-Channel. The arrow indicates the polarity of the PN junction (Arrow points to N).

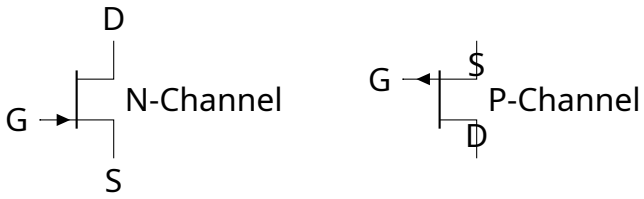


Figure 11.3: Circuit Symbols. Arrow In = N-Channel. Arrow Out = P-Channel.

11.6 Transconductance (g_m)

Since a JFET is a voltage-controlled device, we don't have a current gain (β). Instead, we define **Transconductance (g_m)**: the effectiveness of the Gate voltage in controlling the Drain current.

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{dI_D}{dV_{GS}} \tag{11.2}$$

By differentiating Shockley's equation:

$$g_m = g_{m0}\left(1 - \frac{V_{GS}}{V_P}\right) \tag{11.3}$$

Where g_{m0} is the maximum transconductance (at $V_{GS} = 0$):

$$g_{m0} = \frac{2I_{DSS}}{|V_P|} \tag{11.4}$$

Unit: Siemens (S) or Mhos ($\mho$).

11.7 JFET Biasing (Self-Bias)

The most common biasing method for JFETs is **Self-Bias**. It uses a source resistor R_S to create the necessary reverse bias without a negative power supply.

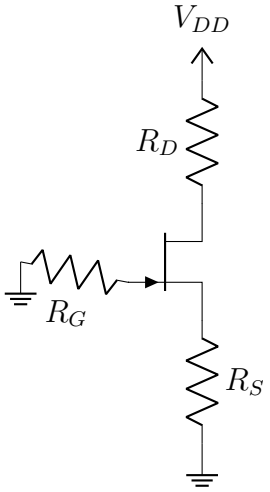


Figure 11.4: Self-Bias Circuit.

Since $I_G = 0$, $V_G = 0$. The source voltage is $V_S = I_D R_S$. Therefore:

$$V_{GS} = V_G - V_S = -I_D R_S \tag{11.5}$$

11.8 Review Questions

11.8.1 Part A: Conceptual Questions

1. Explain the fundamental difference between BJT and JFET in terms of control mechanism and input impedance.
2. Why must the Gate-Source junction of a JFET always be Reverse Biased? What happens if it becomes Forward Biased?
3. Define "Pinch-Off Voltage" (V_P). Does current stop flowing through the channel at Pinch-Off during normal operation?
4. Why are JFETs often preferred over BJTs for the input stages of voltmeters and oscilloscopes?
5. Sketch the construction of a P-Channel JFET and indicate the bias polarities required for operation.
6. What is the physical meaning of I_{DSS}?
7. Explain why the JFET is called a "Unipolar" device.
8. Describe the effect of temperature on JFET drain current compared to BJT collector current.
9. Why is the input current into the Gate of a JFET considered to be zero?
10. Is the JFET construction symmetrical? Can the Source and Drain be swapped?

11.8.2 Part B: Advanced Calculations & Analysis

11. **Shockley Calculation:** A JFET has device parameters $I_{DSS} = 12$ mA and $V_P = -4$ V.
    (a) Calculate the Drain Current (I_D) for $V_{GS} = 0$ V.
    (b) Calculate I_D for $V_{GS} = -2$ V.
    (c) Calculate I_D for $V_{GS} = -6$ V.

12. Bias Point: Determine the value of Gate-Source voltage (V_{GS}) required to bias an N-channel JFET at exactly half of its saturation current ($I_D = I_{DSS}/2$). Given $V_P = -5$ V.

13. **Transconductance:** A JFET has $I_{DSS} = 10$ mA and $V_P = -5$ V.

    (a) Calculate the maximum transconductance g_{m0}.

    (b) Calculate the transconductance g_m at a bias point of $V_{GS} = -2.5$ V.

14. **Self-Bias Analysis:** An N-channel JFET ($I_{DSS} = 8$ mA, $V_P = -4$ V) is used in a self-bias configuration with a source resistor $R_S = 1k\Omega$.

    (a) Write the equation for the self-bias line ($V_{GS} = -I_D R_S$).

    (b) By solving the simultaneous equations (Shockley and Bias Line), determine the Q-point (I_{DQ} and V_{GSQ}).

15. **Design Problem:** You need to design a JFET amplifier stage using a device with $I_{DSS} = 16$ mA and $V_P = -4$ V. Determine the required Source Resistor (R_S) to set the Q-point drain current at 4 mA.